2) Quantum Foundations Refresher

State & measurement. One qubit: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,\ |\alpha|^2 + |\beta|^2 = 1.$

Measure in Z-basis → outcomes $0, 1$ with probs $|\alpha|^2, |\beta|^2$. Post-measurement collapses.

Key single-qubit gates (matrices).

- $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$, $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
- $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ maps $|0\rangle \to |+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$
- $S = \mathrm{diag}(1, i)$, $T = \mathrm{diag}(1, e^{i\pi/4})$

Two-qubit entangler. CNOT(c, t): flips target if control=1.

Global vs. relative phase. $e^{i\phi}|\psi\rangle$ is physically identical to $|\psi\rangle$; but a relative phase (e.g., $|0\rangle + e^{i\theta}|1\rangle$) changes interference.

Practice A (probabilities).

State $|\psi\rangle = \sqrt{3}/2\,|0\rangle + 1/2\,|1\rangle$.

What's P(|0>) after applying a hadamard gate?

Practice B (compose gates).

Apply HZ to $|0\rangle$. Then measure in X-basis.

3) Entanglement & Bell States

Bell prep. Start $|00\rangle$. Apply H on qubit 0, then CX(0→1):
$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. Measuring in Z gives 00 or 11 (50/50), never 01/10.

Reduced state. Each single-qubit reduced density matrix is maximally mixed ($I/2$) even though the two-qubit state is pure → signature of entanglement.

CHSH (what to know). Classically ≤2; quantum can reach $2\sqrt{2}$ using appropriate measurement axes on $|\Phi^+\rangle$. You don't need to derive, just know why entanglement yields stronger correlations.

Teleportation logic (bird's-eye).
(1) Share Bell pair AB. (2) Alice Bell-measures (input, A) and sends 2 classical bits. (3) Bob applies a Pauli correction to B → reconstructs unknown input.

Practice C (detect entanglement).

Given $\rho = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$. Is it entangled?

Practice D (partial trace).

For $|\Phi^+\rangle$, compute $\rho_A = \mathrm{Tr}_B(|\Phi^+\rangle\langle\Phi^+|)$.

4) Quantum Query Algorithms (DJ, BV, Grover)

Oracle model. Black-box $f : \{0,1\}^n \to \{0,1\}$. Standard phase oracle: $|x\rangle \mapsto (-1)^{f(x)}|x\rangle$.

Deutsch–Jozsa (DJ). Promise: f is constant or balanced. One query → measure all-zeros in Hadamard basis iff constant; anything else ⇒ balanced.

Bernstein–Vazirani (BV). Hidden string $s \in \{0,1\}^n$; $f(x) = s \cdot x$ (mod 2). One query with Hadamards before/after reveals s exactly.

Grover (unstructured search). N items, M marked. Iteration = oracle phase flip + diffusion ("about mean"). Optimal $k \approx \left\lfloor \frac{\pi}{4}\sqrt{N/M} \right\rceil$. Over-rotating reduces success.

Practice F (Grover count).

$N = 1024$, $M = 4$. Compute optimal k.

5) Noise & Transpilation Quick Guide

Core noise types (intuition).

- **Amplitude damping (T_1):** excited state $|1\rangle$ relaxes toward $|0\rangle$ over time T_1. Population decay $\sim e^{-t/T_1}$.
- **Dephasing (T_2):** off-diagonal coherence decays $\sim e^{-t/T_2}$ (phase randomization).
- **Depolarizing (p):** state "shrinks" toward maximally mixed: $\rho \mapsto (1-p)\rho + p\frac{I}{2^n}$.
- **Readout error:** classical bit-flip at measurement; model with 2×2 confusion matrices per qubit.

Mitigation playbook (what to write on exam).

- **Readout calibration:** measure $00..0$, $11..1$ to estimate confusion; invert to debias counts.
- **ZNE:** execute with amplified noise (gate folding), fit vs. noise scale → extrapolate to zero.
- **Dynamical decoupling (DD):** idle qubits get refocusing pulses (e.g., X-I-X-I) to suppress low-freq noise.
- **Pauli twirling:** randomize coherent errors to stochastic Pauli channels.

Transpilation levels (Qiskit).

Level 0 (no/very light), 1 (gate merging), 2 (noise-aware layout, basic peephole), 3 (heavier optimizations/commutations). You can set an initial layout to pin logical→physical mapping and reduce SWAPs.

Practice H (T_1/T_2 arithmetic).

A single-qubit idle of $t = 80\,\mathrm{ns}$ on a device with $T_1 = 40\,\mu s$, $T_2 = 20\,\mu s$. Approximate amplitude and phase damping factors?

Practice I (layout thinking).

Your logical CX is between distant physical qubits; level 3 adds 6 SWAPs. What can you do?

6) Phase Estimation (QPE) & Shor Highlights

QPE gist. For unitary $U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle$, use m ancillas to control-apply U^{2^k}, then inverse-QFT; measure m bits ≈ binary expansion of θ (precision $\approx 1/2^m$).

Shor (why factoring reduces to order-finding).

- Pick a coprime to N; define $f(x) = a^x \mod N$.
- Period r of f gives factors via $\gcd(a^{r/2} \pm 1, N)$ (if r even and $a^{r/2} \not\equiv -1 \mod N$).
- QPE on the modular multiplication unitary extracts the phase tied to r; continued fractions turns phase into a rational with denominator r.

Practice J (1-qubit QPE).

Suppose U has eigenphase $\theta = 1/4$ (i.e., eigenvalue $e^{i\pi/2}$). Using **one** ancilla, what's the success chance to read 1 (binary of 0.5)?

- **Solution.** With $m = 1$, resolution is coarse; probabilities are $\cos^2(\pi(\theta - 0))$ vs $\cos^2(\pi(\theta - 1/2))$ after inverse-QFT. Plugging $\theta = 0.25$ → outcome "0" prob = $\cos^2(\pi/4) = 0.5$; "1" also 0.5. Need more ancillas for precision.

Practice K (tiny order finding).

Factor $N = 21$ with $a = 2$. Find the period r of $2^x \bmod 21$.

- **Solution.** $2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16, 2^5 = 11, 2^6 = 1 \mod 21$ ⇒ $r = 6$ (even). Then $a^{r/2} = 2^3 = 8$. $\gcd(8 \pm 1, 21) = \gcd(9, 21) = 3$, $\gcd(7, 21) = 7$. So factors 3 and 7.

7) QAOA (MaxCut example)

Ansatz (p layers).

Start $|+\rangle^{\otimes n}$. For $k = 1..p$: apply $e^{-i\gamma_k H_C}$ then $e^{-i\beta_k H_B}$ with

- **Cost** $H_C = \frac{1}{2}\sum_{(i,j)\in E}(I - Z_i Z_j)$ (counts cut edges),
- **Mixer** $H_B = \sum_i X_i$ (encourages exploration).

  Classical optimizer updates $\{\gamma_k, \beta_k\}$ to minimize $\langle H_C \rangle$.

p=1 intuition. Often good bang-for-buck on sparse graphs; analytic optima known for simple topologies (ring, triangle).

Barren plateaus (what/why). Gradients can vanish with depth/system size; small p, problem-aware initializations, warm-starts help.

Practice L (triangle graph).

Graph: 3 nodes, all connected (triangle). Write H_C.

- **Cost** $H_C = \frac{1}{2}\sum_{(i,j)\in E}(I - Z_i Z_j)$ (counts cut edges),

Practice N (parameter units).

Are γ, β angles or times?

- **Mixer** $H_B = \sum_i X_i$ (encourages exploration).

  Classical optimizer updates $\{\gamma_k, \beta_k\}$ to minimize $\langle H_C \rangle$.

8) Circuit Design Gotchas (and quick fixes)

- **Measuring wrong basis.** To measure in X (Y), rotate: X-basis = apply H; Y-basis = apply $S^\dagger H$; then measure Z.
- **Control/target confusion.** CX(control, target). Always sketch truth table.
- **Ancilla init.** Many algorithms require $|1\rangle$ or $|-\rangle$ ancilla (e.g., DJ, oracles).
- **Endianness & classical bit order.** Qiskit returns bitstrings little-endian by default; label your wires.
- **Over-optimization.** Aggressive transpilation can insert SWAPs; constrain layout or lower level.
- **Reuse without reset.** Measuring then reusing a qubit requires `reset` (otherwise you inherit an unknown post-meas state).

Extra “Know-How in One Breath”

- **Tensor products:** multi-qubit states are Kronecker products; gates acting on subsets are tensored with identity elsewhere.
- **Commutation tips:** $Z_i Z_j$ commute with each other; mixers X_i commute across different qubits; use this to simplify exponentials.
- **Grover math:** success $\sin^2((2k+1)\theta)$ with $\sin^2\theta = M/N$; choose k to push angle near $\pi/2$.
- **QPE precision:** m ancillas ≈ m bits of phase; success improves with phase kickback powers U^{2^k}.

Handy Pauli cheat sheet (1-qubit)

Squares and identity

- $X^2=Y^2=Z^2=I$
- $T^4=S^2=Z$, $T^2=S$

Products (remember order matters)

- XY=iZ
- YZ=iX
- ZX=iY
- YX=-iZ
- ZY=-iX
- XZ=-iY

(Equivalently: swapping two different Paulis flips the sign and complex phase: PQ=-QP for P≠QP)

Conjugation by a Pauli

- PQP={ Q if P and Q commute {X,X}, {Y,Y}, and {Z,Z} commute }
  {-Q if P and Q anticommute, {X,Y}, {Y,Z}, and {Z,X} anticommute}
- So
  - XZX=-Z,
  - XYX=-Y,
  - but XXX=X.

Useful basis-change (non-Pauli) identities (often show up in reductions)

- HXH=Z,
- HZH=X,
- HYH=-Y
- SXS†=Y,
- SYS†=-X,
- SZS†=Z

1. **Gate algebra.** Show $HZH = X$ and $HXH = Z$.

2. **Bell correlations.** For $|\Phi^+\rangle$, what is $\langle Z \otimes Z \rangle$ and $\langle X \otimes X \rangle$?

3. **Grover multi-solution.** $N = 2048$, $M = 8$. Optimal k?